

CSE 311:

Data Communication

Instructor:
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Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Fourier Series of a periodic signal

‘Any function that *periodically repeats itself* can be expressed as the *sum of sines and/or cosines* of different frequencies’

- Joseph Fourier

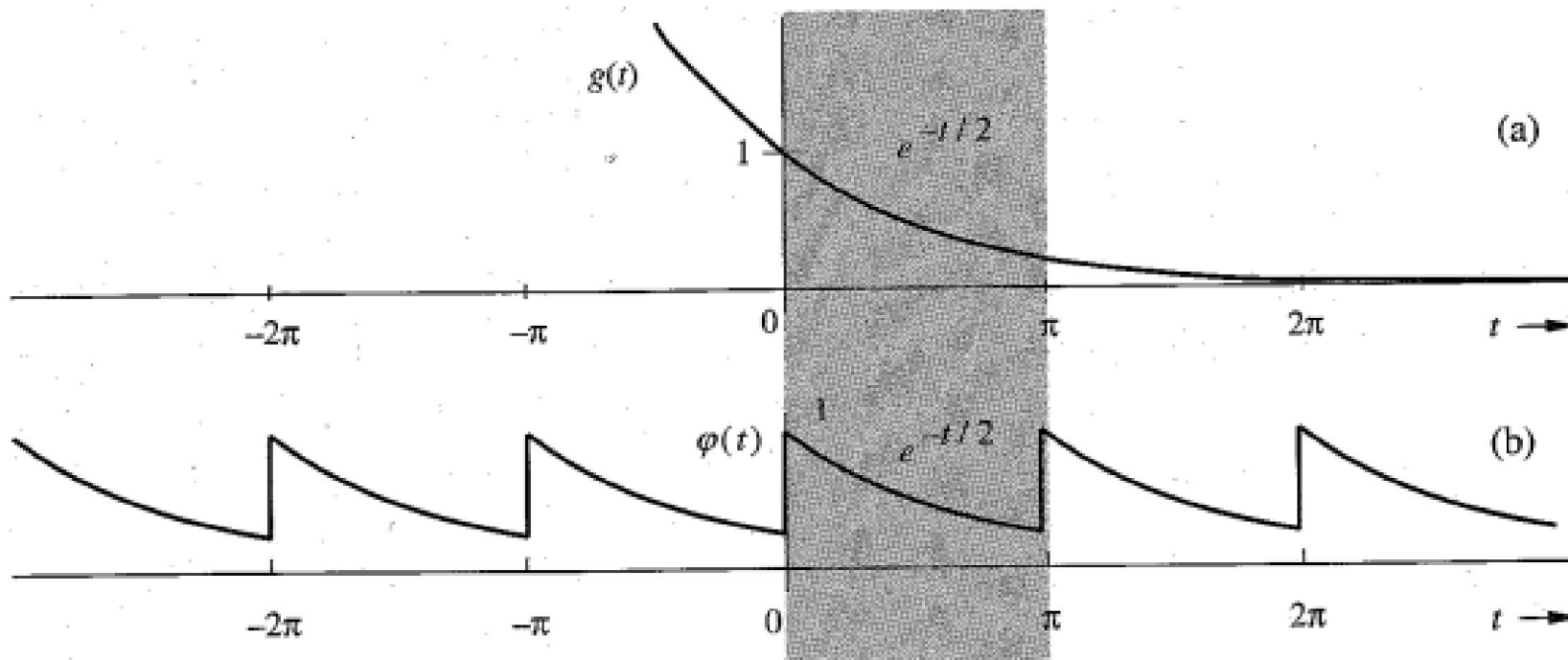
$$g(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty}^{n=\infty} D_n e^{jn2\pi f_0 t}$$

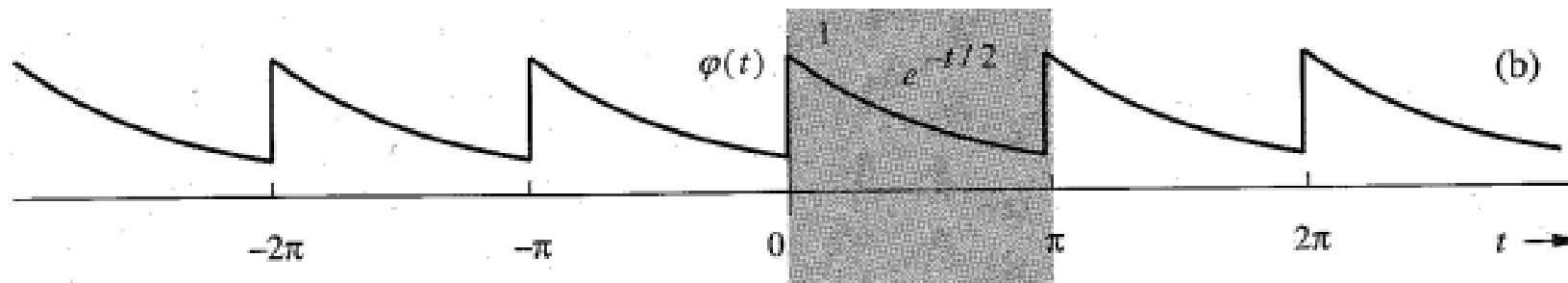
with, T_0 = period of $g(t)$
 f_0 = frequency of $g(t)$

$$\text{where, } D_n = \frac{1}{T_0} \int_{T_0} g(t) e^{-jn2\pi f_0 t}$$

Representing a Periodic signal using Fourier Series



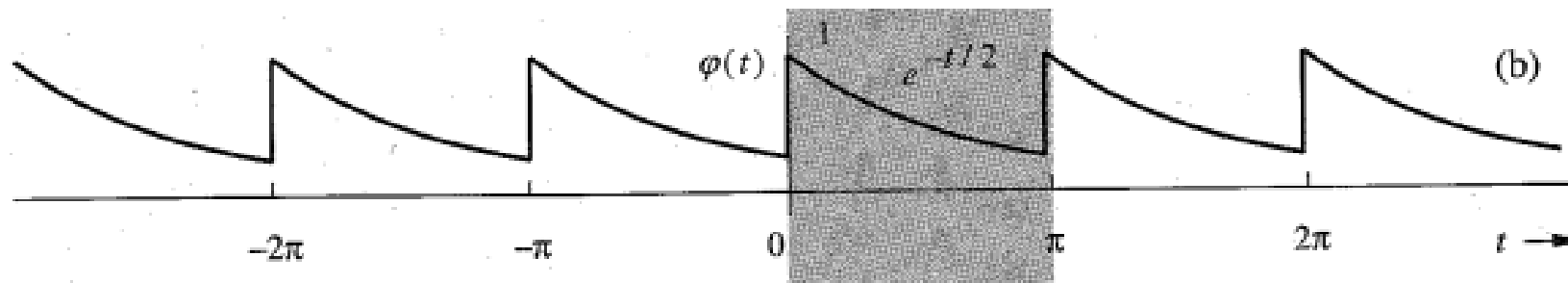
Representing a Periodic signal using Fourier Series



$$\phi(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{n=\infty} D_n e^{j2nt}$$

Here, $T_0 = \pi$ and $\omega_0 = \frac{2\pi}{T_0} = 2 \text{ rad/s}$

Representing a Periodic signal using Fourier Series

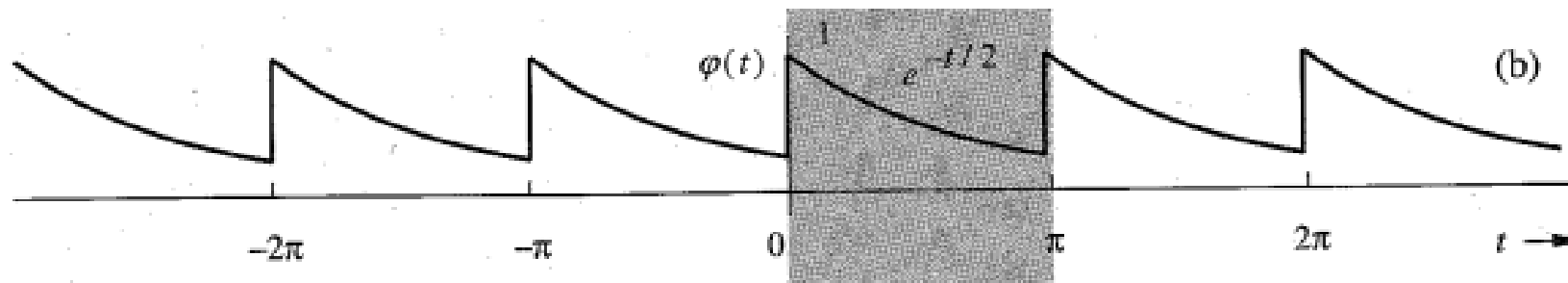


$$D_n = \frac{1}{T_0} \int_{T_0} \varphi(t) e^{-j2nt} dt = \frac{1}{\pi} \int_0^{\pi} e^{-t/2} e^{-j2nt} dt$$

$$= \frac{1}{\pi} \int_0^{\pi} e^{-(\frac{1}{2} + j2n)t} dt = \frac{-1}{\pi (\frac{1}{2} + j2n)} e^{-(\frac{1}{2} + j2n)t} \Big|_0^{\pi}$$

$$= \frac{0.504}{1 + j4n}$$

Representing a Periodic signal using Fourier Series



$$D_n = \frac{0.504}{1 + j4n}$$

$$\phi(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{j2nt}$$

$$\phi(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2nt}$$

$$= 0.504 \left[1 + \frac{1}{1 + j4} e^{j2t} + \frac{1}{1 + j8} e^{j4t} + \frac{1}{1 + j12} e^{j6t} + \dots \right. \\ \left. + \frac{1}{1 - j4} e^{-j2t} + \frac{1}{1 - j8} e^{-j4t} + \frac{1}{1 - j12} e^{-j6t} + \dots \right]$$

Representing a Periodic signal using Fourier Series

- D_n are exponential Fourier spectra
- D_n are complex though $\phi(t)$ is a real periodic
- D_n and D_{-n} are complex conjugates

$$D_n = \frac{0.504}{1 + j4n}$$

$$\varphi(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2nt}$$

$$= 0.504 \left[1 + \frac{1}{1 + j4} e^{j2t} + \frac{1}{1 + j8} e^{j4t} + \frac{1}{1 + j12} e^{j6t} + \dots \right. \\ \left. + \frac{1}{1 - j4} e^{-j2t} + \frac{1}{1 - j8} e^{-j4t} + \frac{1}{1 - j12} e^{-j6t} + \dots \right]$$

Representing a Periodic signal using Fourier Series

- D_n are complex though $\phi(t)$ is a **real** periodic
 - $D_n = |D_n|e^{j\angle D_n}$

Representing a Periodic signal using Fourier Series

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 - $D_n = |D_n|e^{j\angle D_n}$
- D_n and D_{-n} are complex conjugates
 - $|D_n| = |D_{-n}|$
 - $\angle D_n = \theta_n$ and $\angle D_{-n} = -\theta_n$

Representing a Periodic signal using Fourier Series

- D_n are complex though $\phi(t)$ is a **real** periodic
 - $D_n = |D_n|e^{j\angle D_n}$
- D_n and D_{-n} are complex conjugates
 - $|D_n| = |D_{-n}|$
 - $\angle D_n = \theta_n$ and $\angle D_{-n} = -\theta_n$
- Therefore, $D_n = |D_n|e^{j\theta_n}$ and $D_{-n} = |D_n|e^{-j\theta_n}$

Representing a Periodic signal using Fourier Series

$$|D_n| = |D_{-n}|$$

$$\angle D_n = \theta_n \quad \text{and} \quad \angle D_{-n} = -\theta_n$$

- $|D_n|$ are amplitudes and $\angle D_n$ are angles of exponential spectra
- $|D_n|$ vs. f is an even function
- $\angle D_n$ vs. f is an odd function

Representing a Periodic signal using Fourier Series

$$D_n = \frac{0.504}{1 + j4n}$$

$$D_0 = 0.504$$

$$D_1 = \frac{0.504}{1 + j4} = 0.122e^{-j75.96^\circ} \implies |D_1| = 0.122 \quad \angle D_1 = -75.96^\circ$$

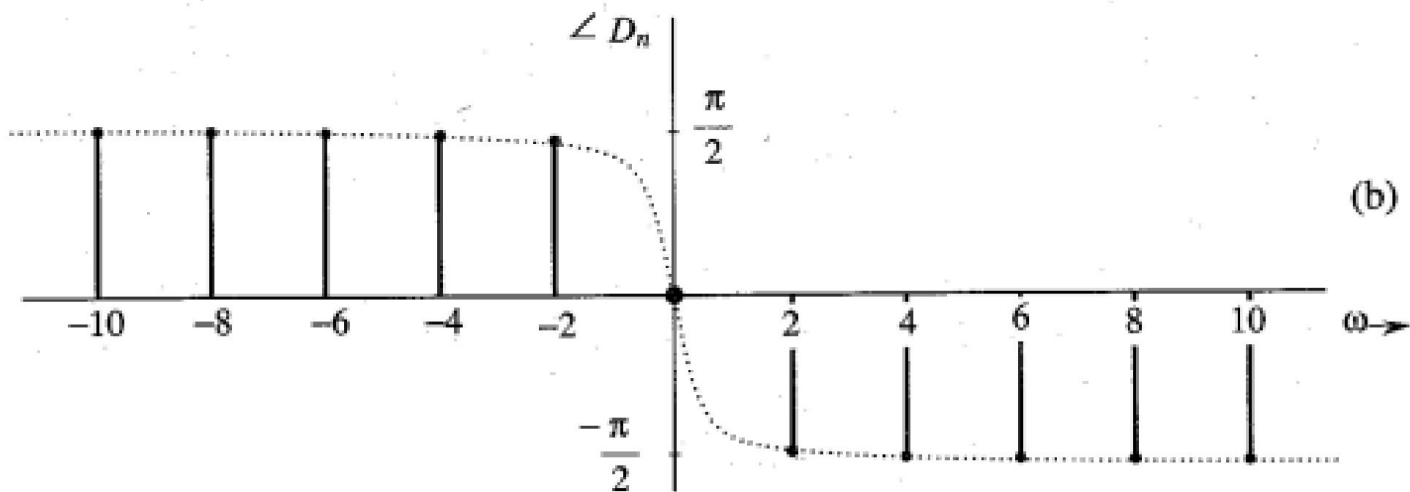
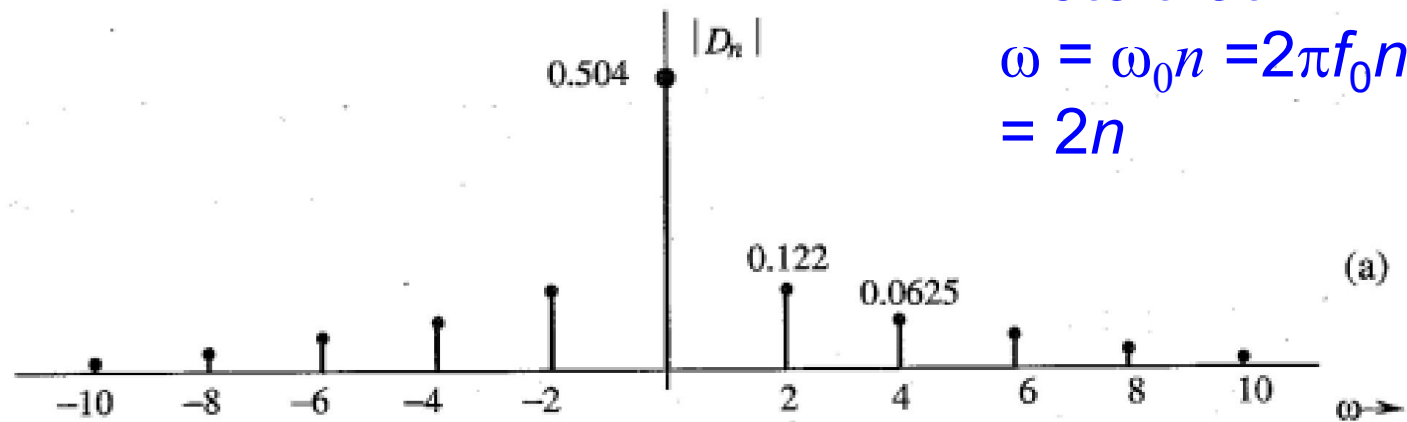
$$D_{-1} = \frac{0.504}{1 - j4} = 0.122e^{j75.96^\circ} \implies |D_{-1}| = 0.122 \quad \angle D_{-1} = 75.96^\circ$$

$$D_2 = \frac{0.504}{1 + j8} = 0.0625e^{-j82.87^\circ} \implies |D_2| = 0.0625 \quad \angle D_2 = -82.87^\circ$$

$$D_{-2} = \frac{0.504}{1 - j8} = 0.0625e^{j82.87^\circ} \implies |D_{-2}| = 0.0625 \quad \angle D_{-2} = 82.87^\circ$$

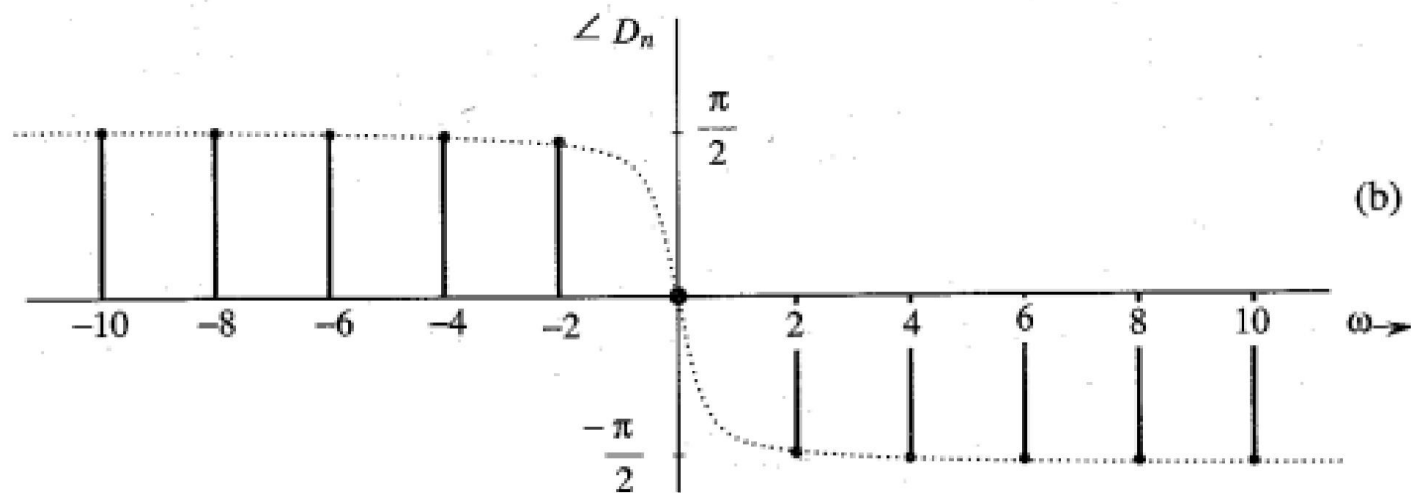
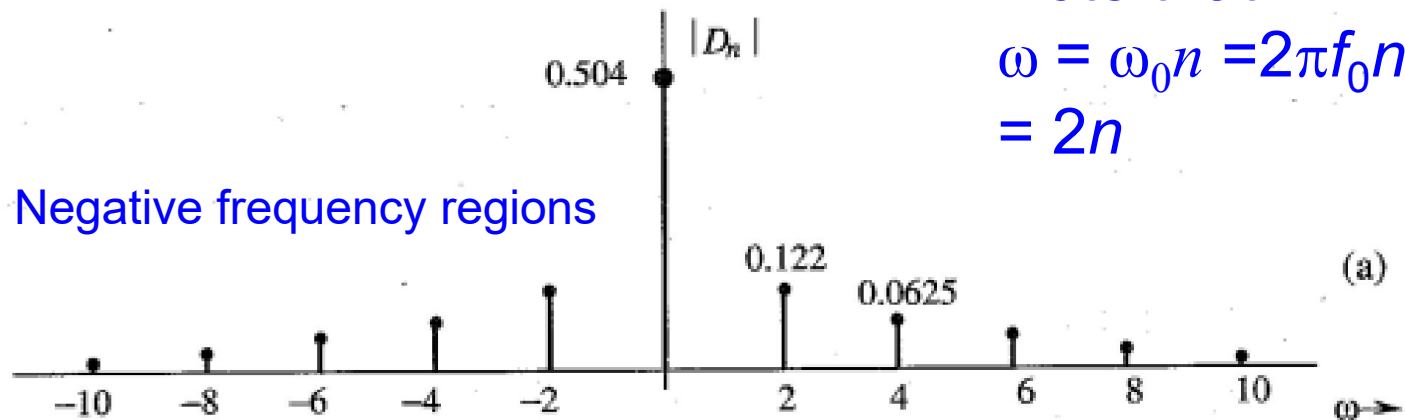
Representing a Periodic signal using Fourier Series

Note that
 $\omega = \omega_0 n = 2\pi f_0 n$
 $= 2n$



Representing a Periodic signal using Fourier Series

Note that
 $\omega = \omega_0 n = 2\pi f_0 n$
 $= 2n$



Meaning of negative frequencies

Consider this real periodic function with frequency f_0

$$\cos(-\omega_0 t + \theta) = \cos(\omega_0 t - \theta)$$

Note that
angular
frequency is
 $\omega_0 = 2\pi f_0$

Meaning of negative frequencies

Consider this real periodic function with frequency f_0

$$\cos(-\omega_0 t + \theta) = \cos(\omega_0 t - \theta)$$

Here, angular frequency is $|\omega_0|$, a positive quantity which only indicates rate of sinusoidal variation

Meaning of negative frequencies

Consider this real periodic function with frequency f_0

$$\cos(-\omega_0 t + \theta) = \cos(\omega_0 t - \theta)$$

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Cannot explain negative frequency

Meaning of negative frequencies

Consider this **complex** sinusoid function with frequency f_0

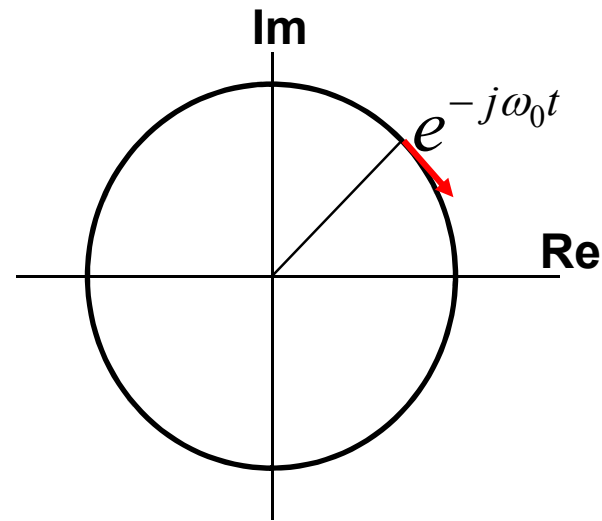
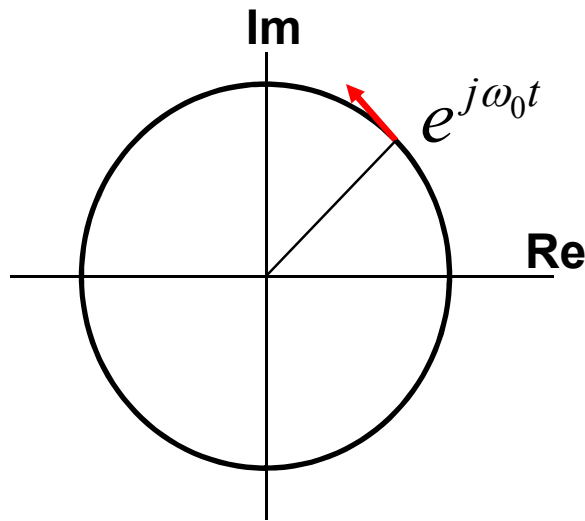
$$e^{\pm j\omega_0 t} = \cos \omega_0 t \pm j \sin \omega_0 t$$

It both describes the **rate variation**, $|\omega_0|$, and **direction** ($\pm$)

Meaning of negative frequencies

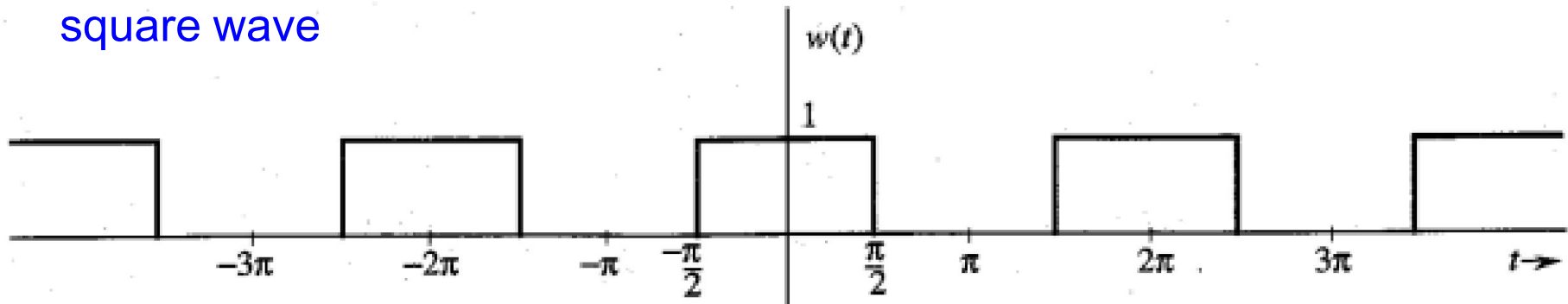
Consider this **complex** sinusoid
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$$e^{\pm j\omega_0 t} = \cos \omega_0 t \pm j \sin \omega_0 t$$



Fourier series of Some Useful Signals

Periodic
square wave

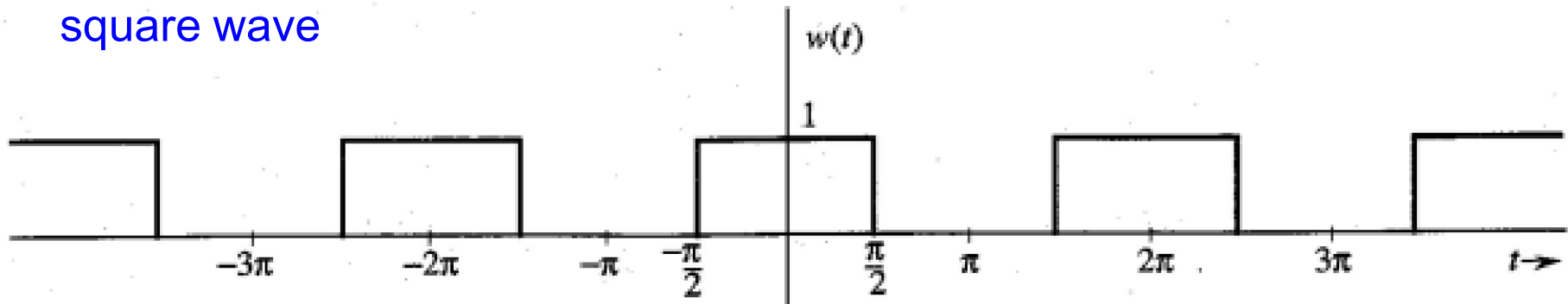


Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

Fourier series of Some Useful Signals

Periodic
square wave



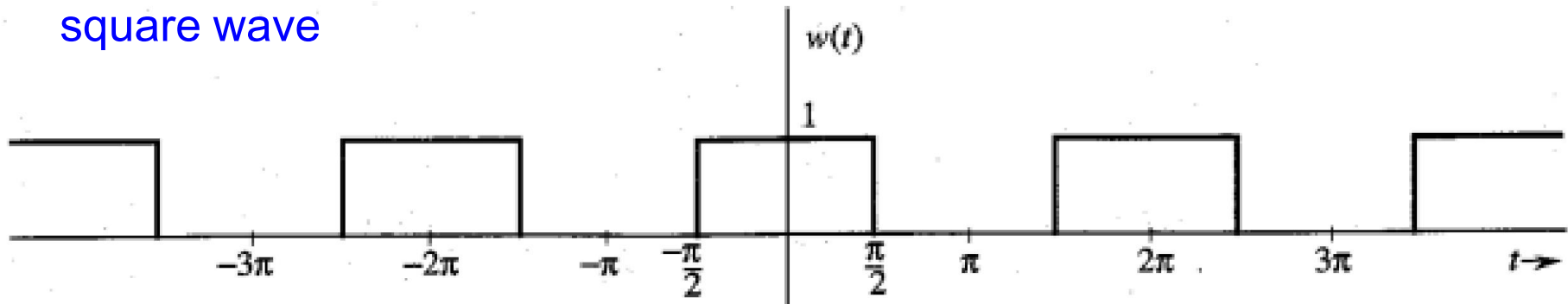
Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$w(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \quad \text{where, } \omega_0 = 2\pi f_0$$

Fourier series of Some Useful Signals

Periodic
square wave



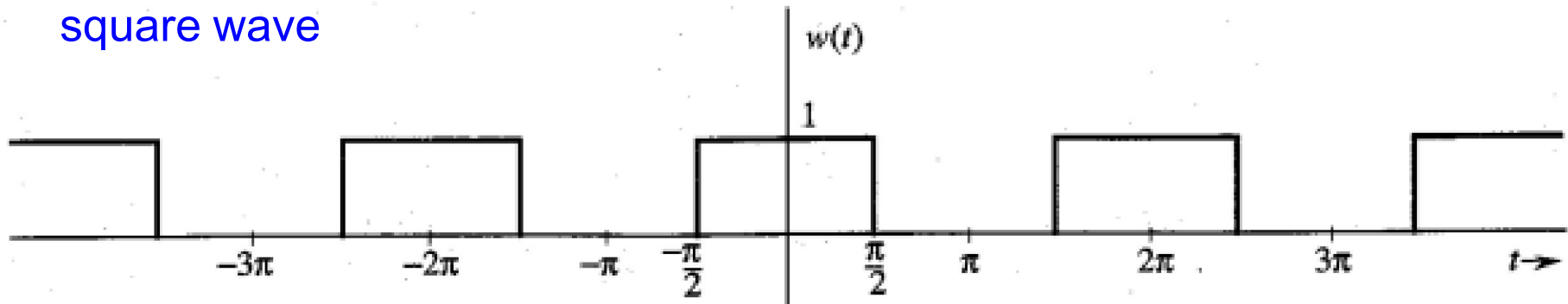
Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$D_n = \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t} dt$$

Fourier series of Some Useful Signals

Periodic
square wave



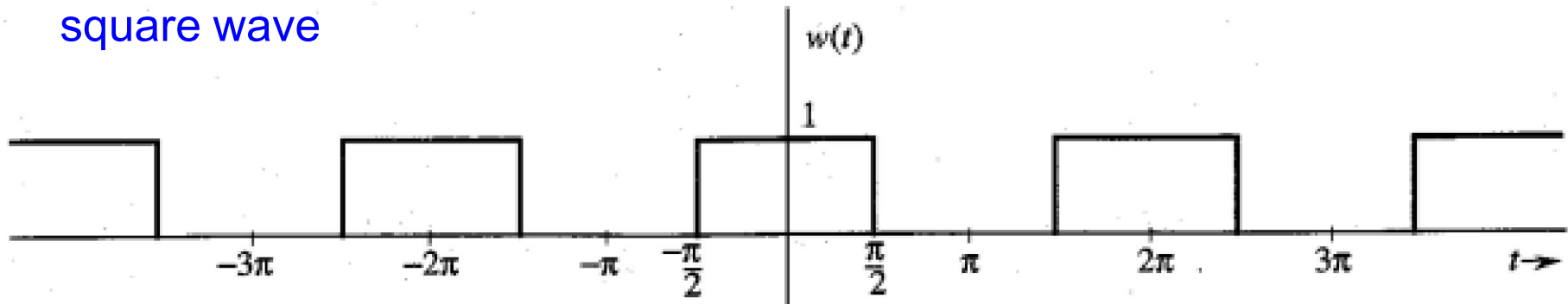
Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$D_n = \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t} dt$$

Fourier series of Some Useful Signals

Periodic
square wave



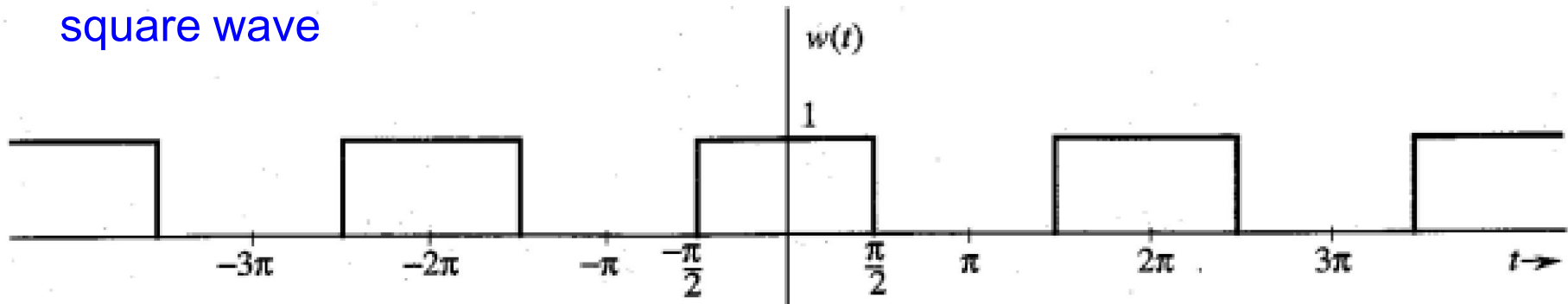
Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$\begin{aligned} D_n &= \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t} dt \\ &= \frac{1}{-jn\omega_0 T_0} (e^{-jn\omega_0 T_0/4} - e^{jn\omega_0 T_0/4}) \end{aligned}$$

Fourier series of Some Useful Signals

Periodic
square wave



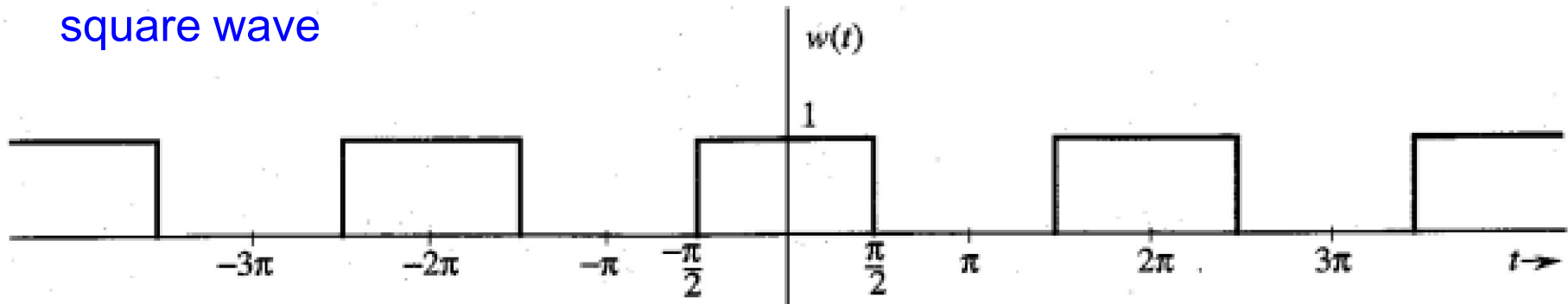
Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$\begin{aligned} D_n &= \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t} dt \\ &= \frac{1}{-jn\omega_0 T_0} (e^{-jn\omega_0 T_0/4} - e^{jn\omega_0 T_0/4}) \\ &= \frac{2}{n\omega_0 T_0} \sin\left(\frac{n\omega_0 T_0}{4}\right) = \frac{1}{n\pi} \sin\left(\frac{n\pi}{2}\right) \end{aligned}$$

Fourier series of Some Useful Signals

Periodic
square wave



Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

$$D_n = \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0} \int_{-T_0/4}^{T_0/4} e^{-jn\omega_0 t} dt$$

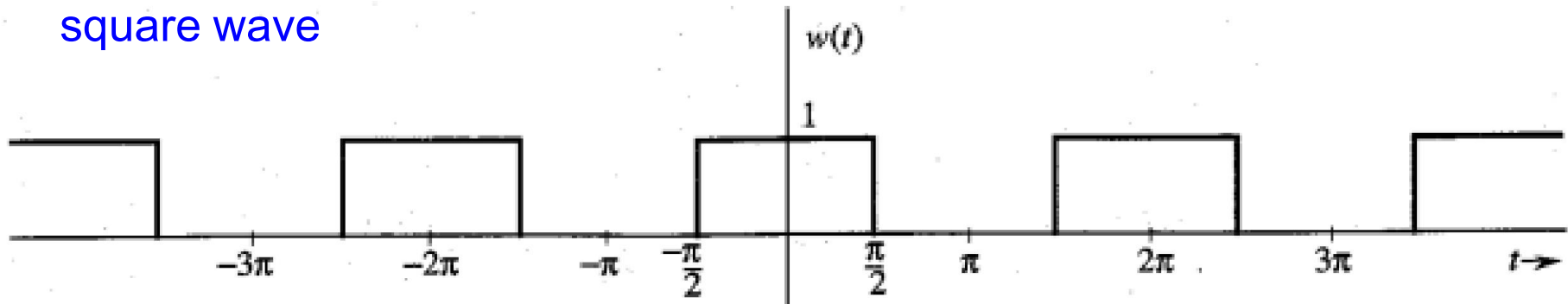
$$= \frac{1}{-jn\omega_0 T_0} (e^{-jn\omega_0 T_0/4} - e^{jn\omega_0 T_0/4})$$

$$= \frac{2}{n\omega_0 T_0} \sin\left(\frac{n\omega_0 T_0}{4}\right) = \frac{1}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

D_0 is undefined

Fourier series of Some Useful Signals

Periodic
square wave



Period, $T_0 = 2\pi$

Frequency, $f_0 = 1/T_0 = 1/2\pi$

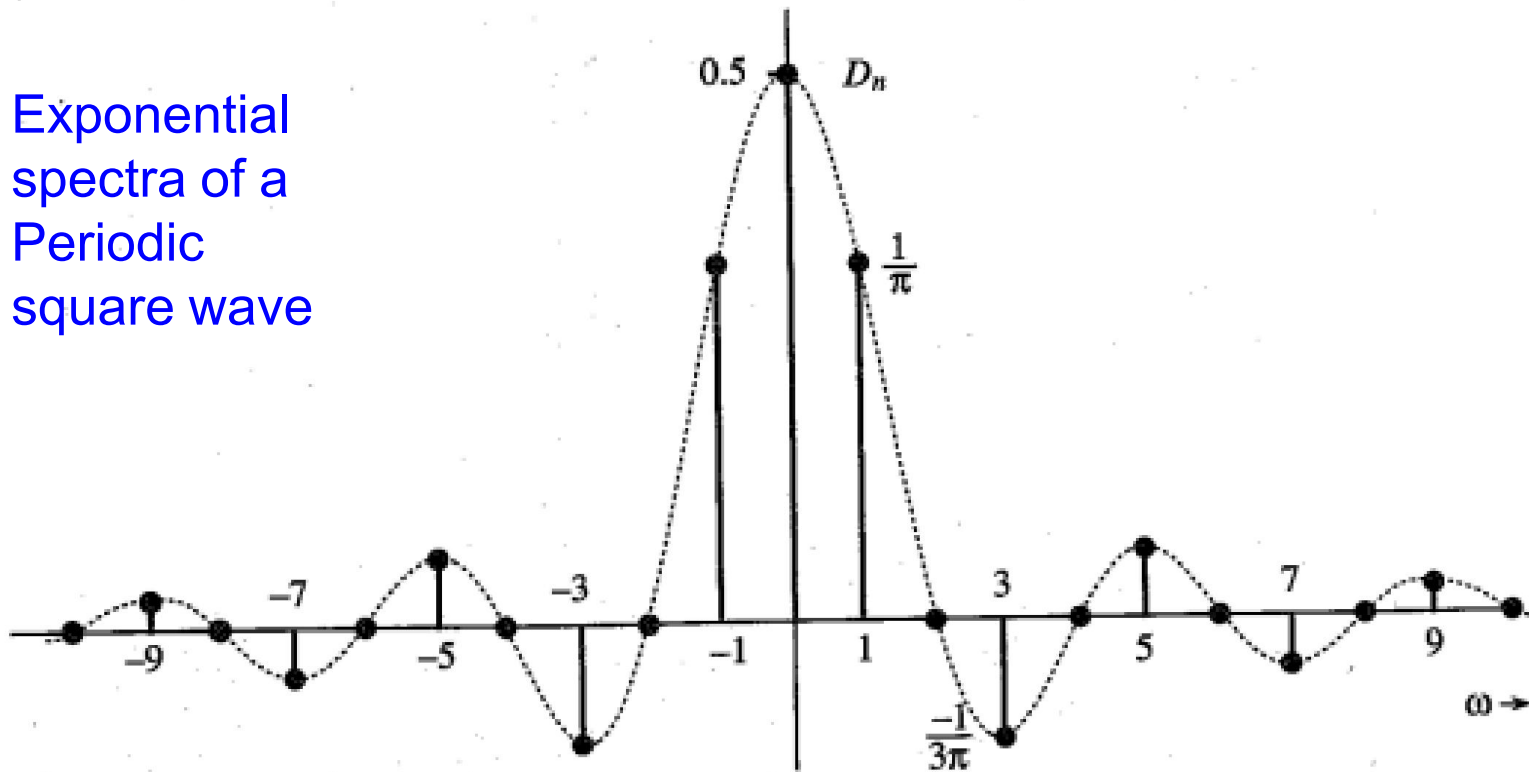
$$D_n = \frac{1}{T_0} \int_{T_0} w(t) e^{-jn\omega_0 t} dt = \frac{1}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

$$D_0 = \frac{1}{T_0} \int_{T_0} w(t) dt = \frac{1}{2}$$

Fourier series of Some Useful Signals

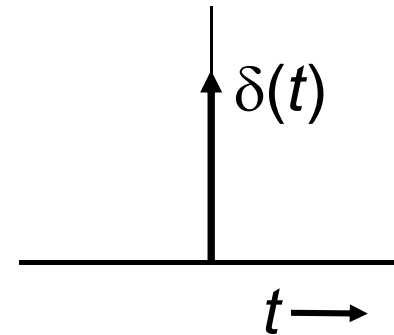
$$D_n = \frac{1}{n\pi} \sin\left(\frac{n\pi}{2}\right) \quad D_0 = \frac{1}{2}$$

Exponential
spectra of a
Periodic
square wave

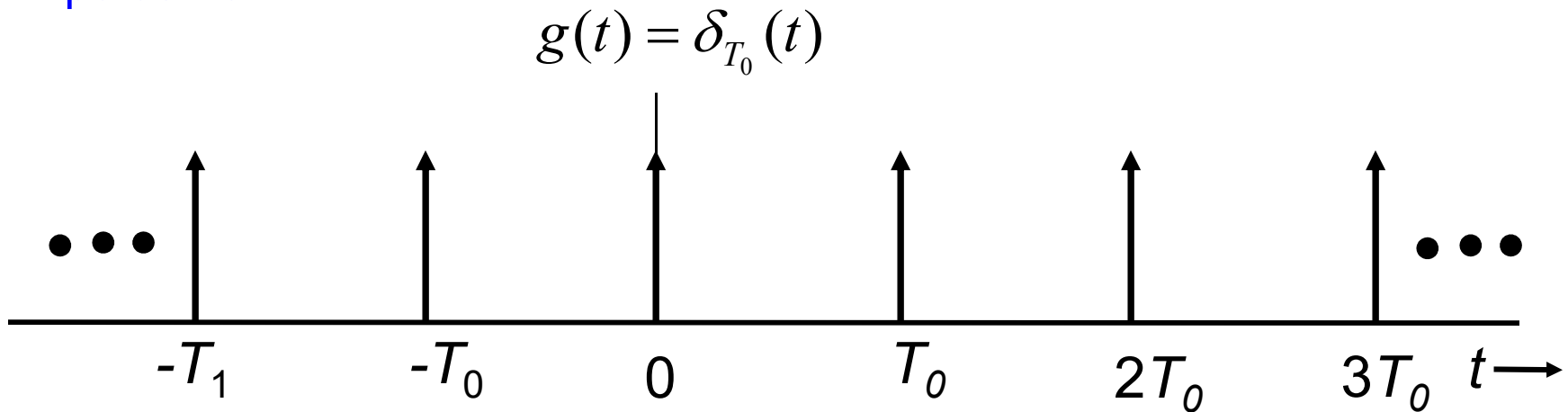


Fourier Series of Unit Impulse Train

Impulse signal

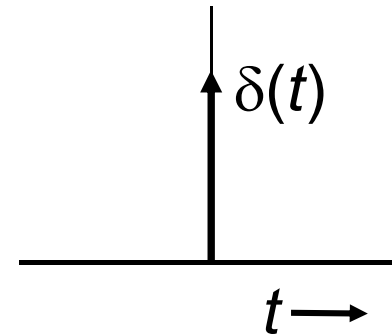


Impulse train

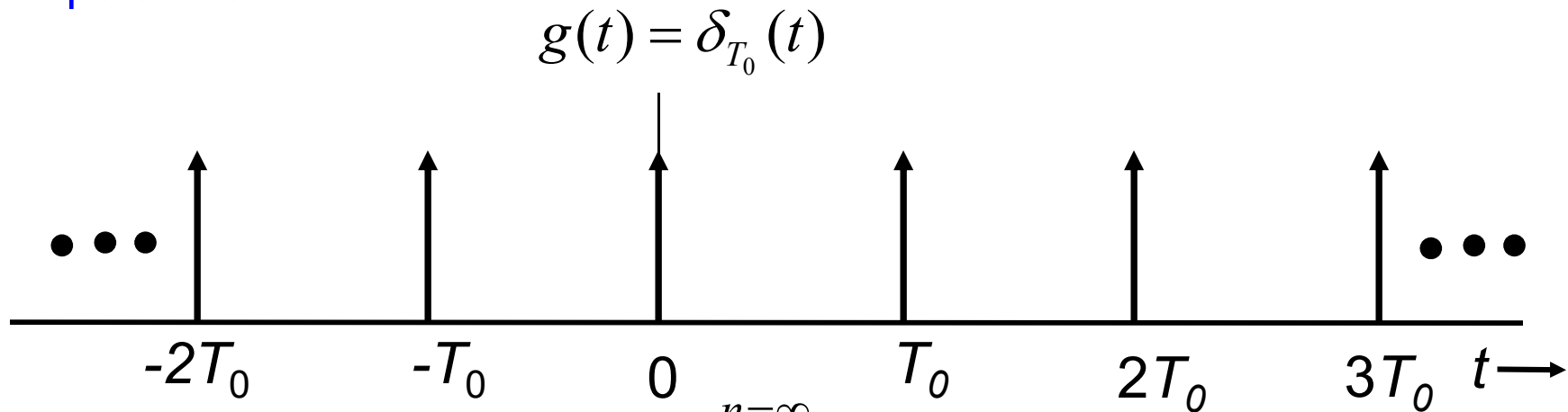


Fourier Series of Unit Impulse Train

Impulse signal



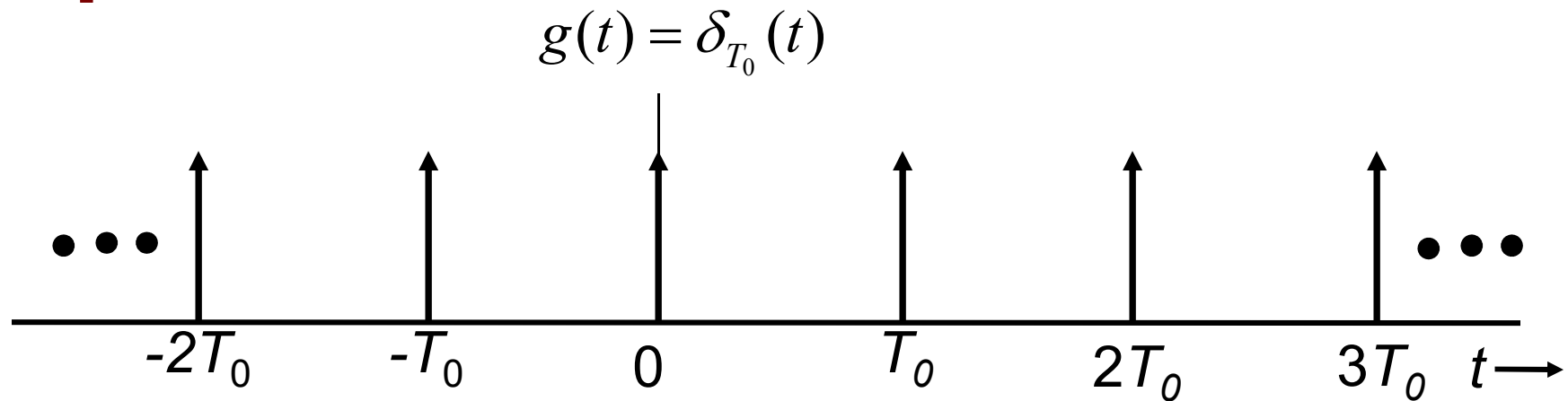
Impulse train



Equation of an Impulse train

$$s_{T_0}(t) = \sum_{n=-\infty}^{n=\infty} \delta(t - nT_0)$$

Fourier Series of Unit Impulse Train

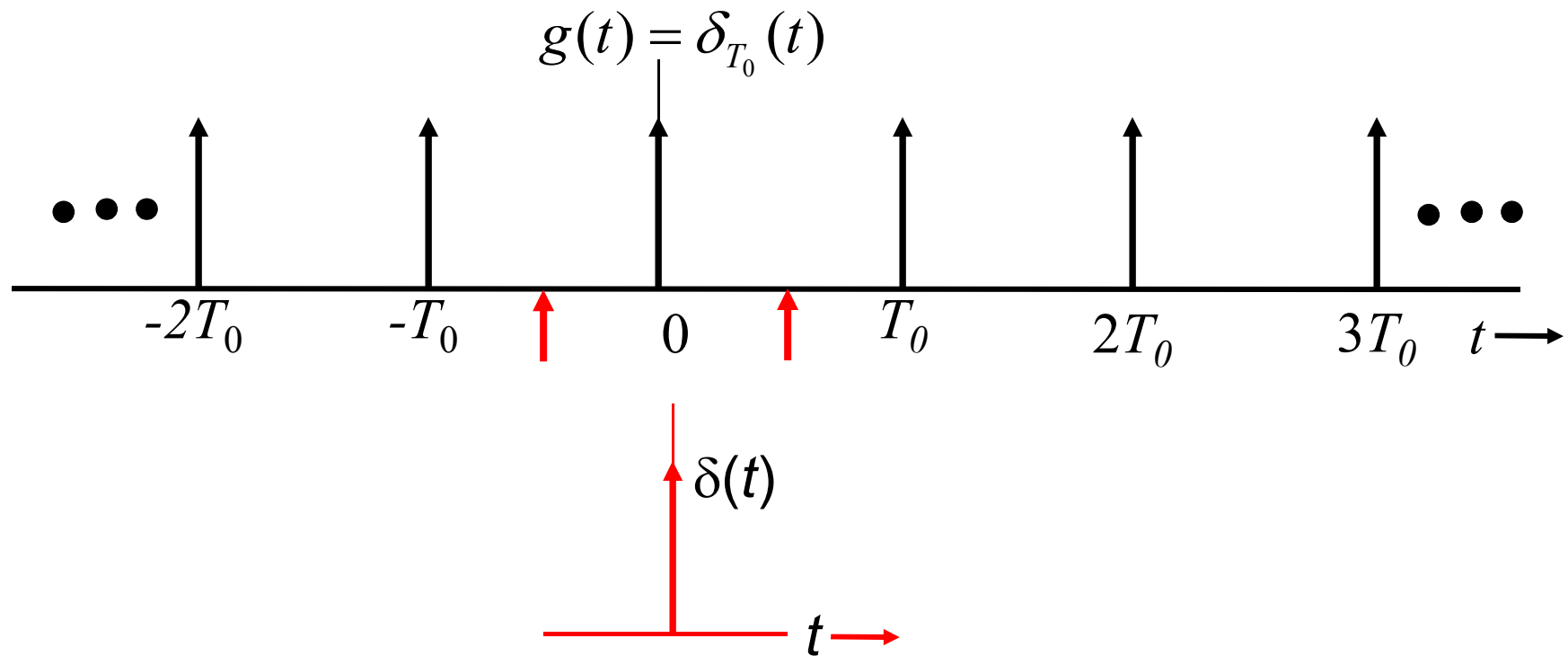


As it is a periodic signal with period T_0

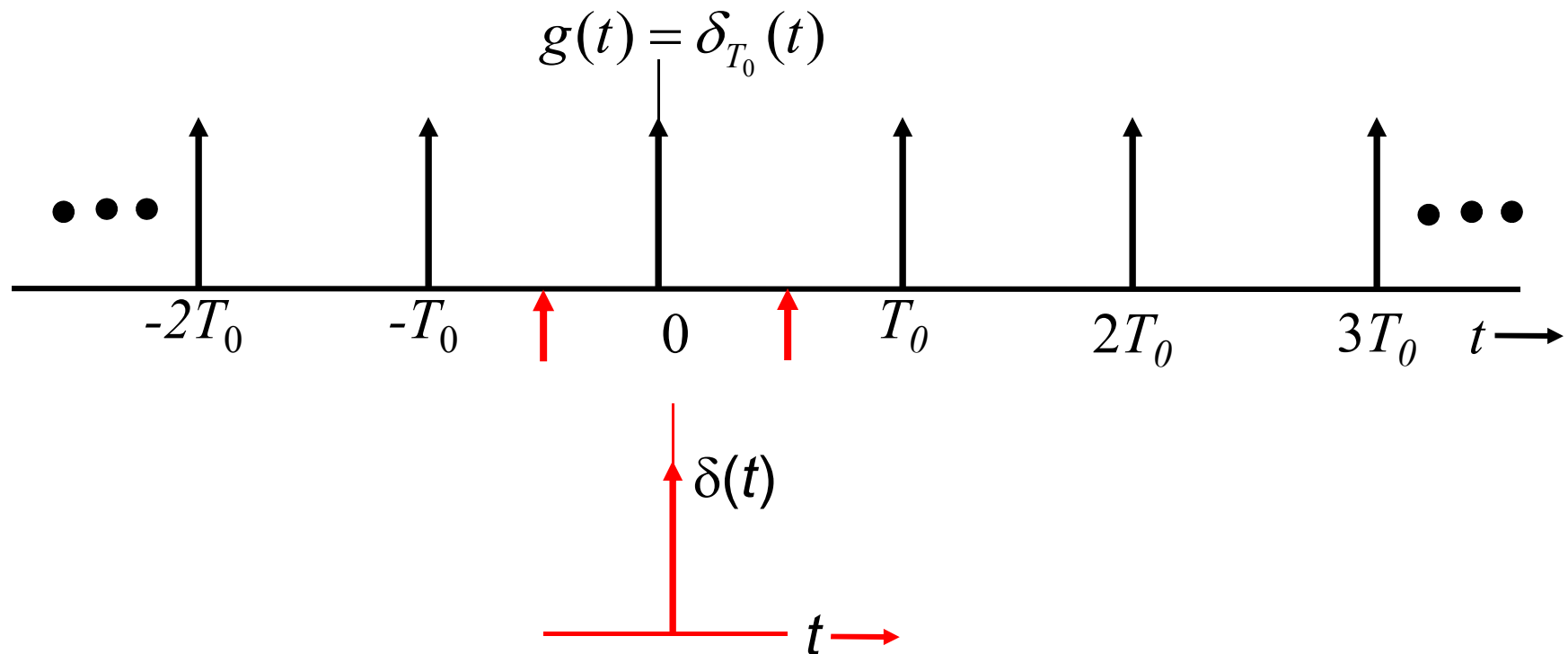
$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} \quad \omega_0 = \frac{2\pi}{T_0}$$

where, $D_n = \frac{1}{T_0} \int_{T_0} \delta_{T_0}(t) e^{-jn\omega_0 t} dt$

Fourier Series of Unit Impulse Train

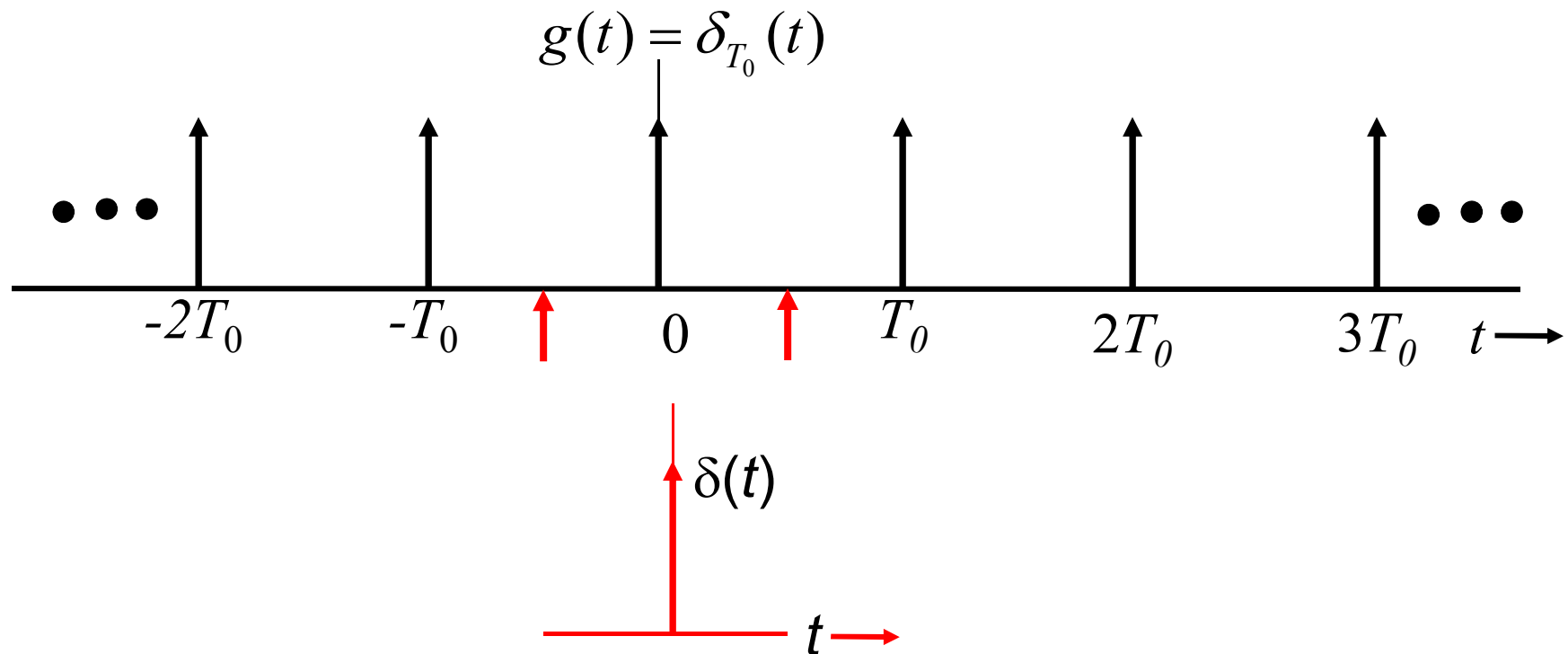


Fourier Series of Unit Impulse Train



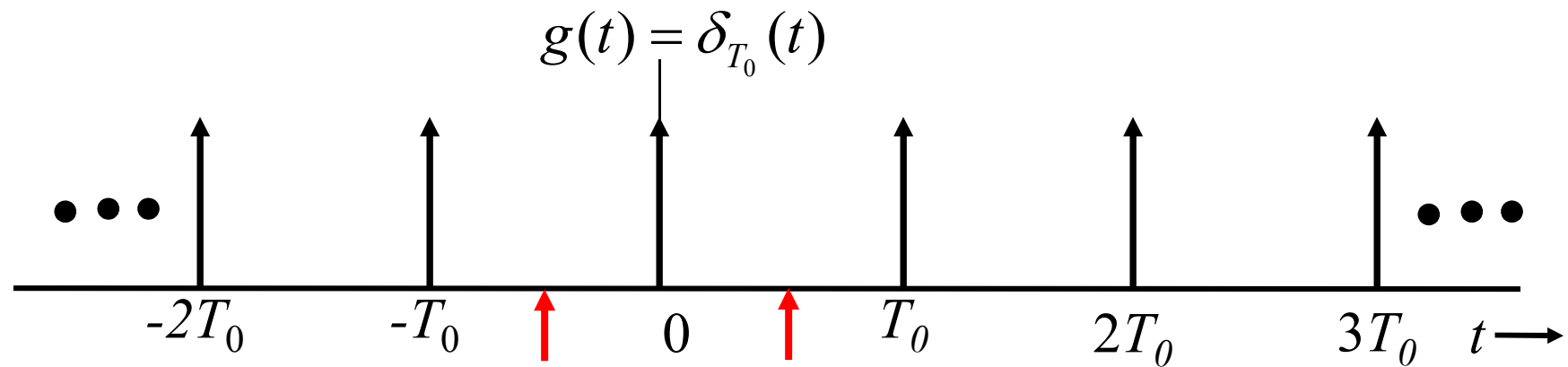
$$D_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t) e^{-jn\omega_0 t} dt$$

Fourier Series of Unit Impulse Train

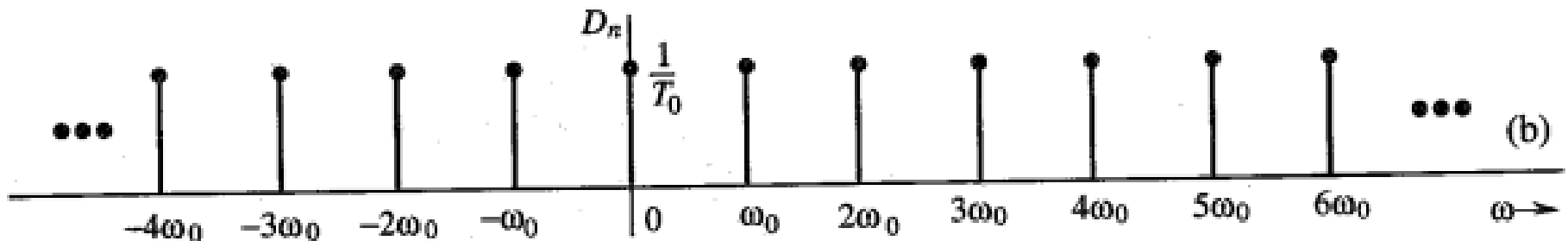


$$D_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} \delta(t) e^{-jn\omega_0 t} dt = \frac{1}{T_0}$$

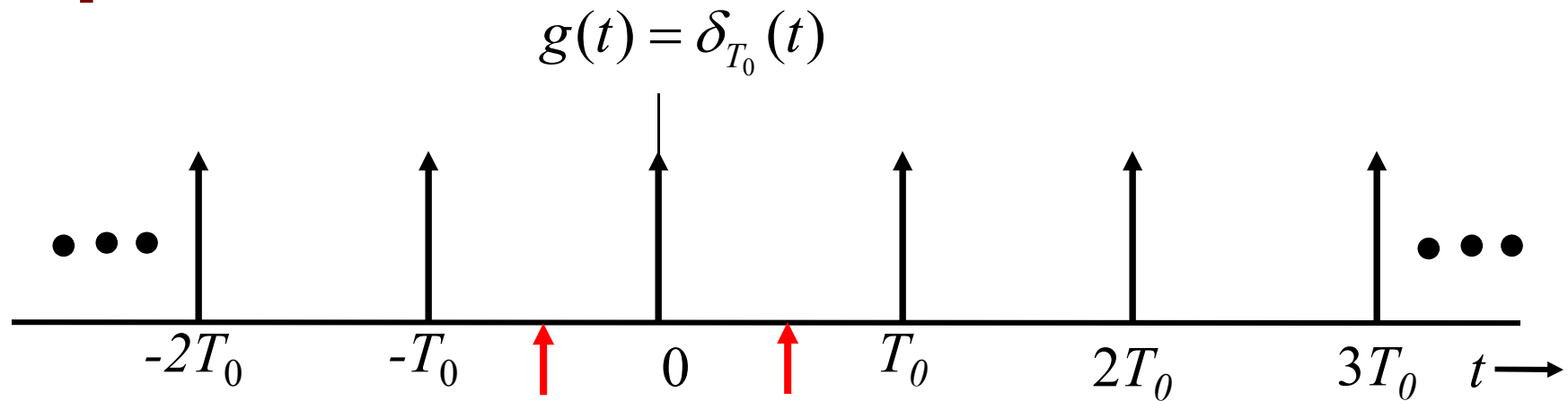
Fourier Series of Unit Impulse Train



$$D_n = \frac{1}{T_0}$$



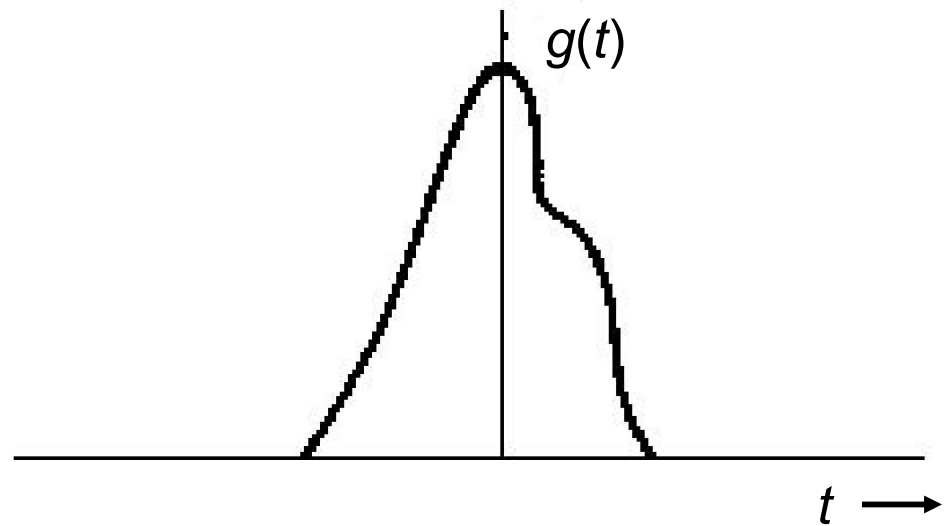
Fourier Series of Unit Impulse Train



Fourier Series is

$$\delta_{T_0}(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t} = \frac{1}{T_0} \sum_{n=-\infty}^{\infty} e^{jn\omega_0 t}$$

Fourier's theorem for aperiodic Signal



Fourier's theorem for aperiodic Signal

- *aperiodic* signal but *with finite area under curve* can be represented as *integral* of sines and cosines

$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df$$

$$\text{where, } G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt$$

$$\text{and } \omega = 2\pi f$$

Fourier's theorem for aperiodic Signal

- These two are called Fourier transform pair

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt$$
$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df = \int_{-\infty}^{\infty} G(f) e^{j2\pi f t} df$$

Fourier's theorem for aperiodic Signal

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$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$
$$g(t) = \int_{-\infty}^{\infty} G(f) e^{j\omega t} df = \int_{-\infty}^{\infty} G(f) e^{j2\pi ft} df$$

Symbolically shown as,

$$g(t) \Leftrightarrow G(f) \text{ or } G(f) = \mathfrak{F}[g(t)]$$

Fourier Integral for aperiodic signal

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt$$

- $G(f)$ is Fourier spectra and complex even if $g(t)$ is real
- $G(f)$ can be shown as

$$G(f) = |G(f)| e^{j\theta_g(f)}$$

Fourier Integral for aperiodic signal

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft} dt$$

- Replacing f by $-f$

$$G(-f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt$$

Fourier Integral for aperiodic signal

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft} dt$$

- Replacing f by $-f$

$$G(-f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt$$

$$G^*(f) = \int_{-\infty}^{\infty} g(t)e^{j2\pi ft} dt \quad \text{provided, } g(t) \text{ is real}$$

Fourier Integral for aperiodic signal

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j\omega t} dt = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt$$

- Replacing f by $-f$

$$G(-f) = \int_{-\infty}^{\infty} g(t) e^{j2\pi f t} dt$$

$$G^*(f) = \int_{-\infty}^{\infty} g(t) e^{j2\pi f t} dt = G(-f)$$

Fourier Integral for aperiodic signal

- The following is complex conjugate symmetry

$$G^*(f) = G(-f)$$

Fourier Integral for aperiodic signal

- The following is complex conjugate symmetry

$$G^*(f) = G(-f)$$

Therefore,

$$|G(f)| = |G^*(f)| = |G(-f)|$$

Fourier Integral for aperiodic signal

- The following is complex conjugate symmetry

$$G^*(f) = G(-f)$$

Therefore,

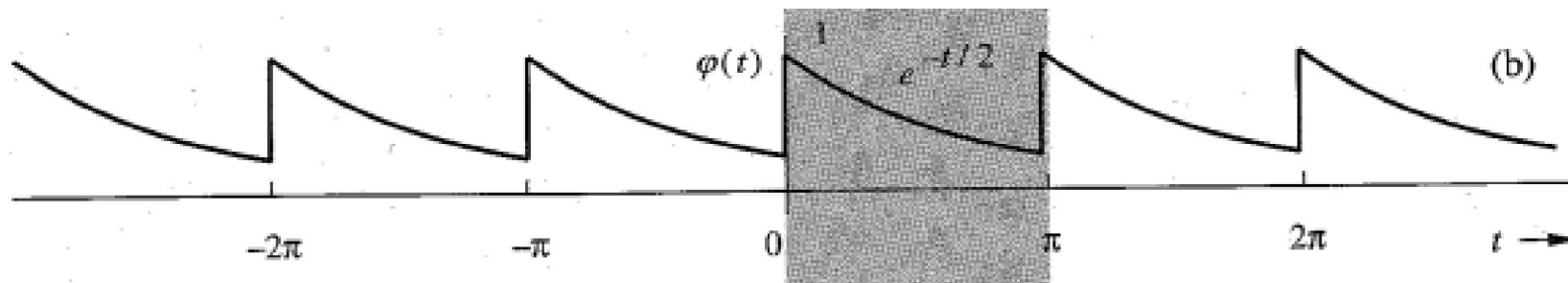
$$|G(f)| = |G^*(f)| = |G(-f)|$$

but

$$\theta_g(-f) = -\theta_g(f)$$

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited!



$$\phi(t) = \sum_{n=-\infty}^{n=\infty} D_n e^{jn\omega_0 t} = \sum_{n=-\infty}^{n=\infty} D_n e^{j2nt}$$

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited!

- D_n are exponential Fourier spectra
- D_n are complex though $\phi(t)$ is a real periodic
- D_n and D_{-n} are complex conjugates

$$D_n = \frac{0.504}{1 + j4n}$$

$$\varphi(t) = 0.504 \sum_{n=-\infty}^{\infty} \frac{1}{1 + j4n} e^{j2nt}$$

$$\begin{aligned} &= 0.504 \left[1 + \frac{1}{1 + j4} e^{j2t} + \frac{1}{1 + j8} e^{j4t} + \frac{1}{1 + j12} e^{j6t} + \dots \right. \\ &\quad \left. + \frac{1}{1 - j4} e^{-j2t} + \frac{1}{1 - j8} e^{-j4t} + \frac{1}{1 - j12} e^{-j6t} + \dots \right] \end{aligned}$$

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited

- D_n are complex though $\phi(t)$ is a **real** periodic
 - $D_n = |D_n|e^{j\angle D_n}$
- D_n and D_{-n} are complex conjugates
 - $|D_n| = |D_{-n}|$
 - $\angle D_n = \theta_n$ and $\angle D_{-n} = -\theta_n$
- Therefore, $D_n = |D_n|e^{j\theta_n}$ and $D_{-n} = |D_n|e^{-j\theta_n}$

Conjugate Symmetry Property in Fourier Series of Periodic signals

Revisited!

$$|D_n| = |D_{-n}|$$

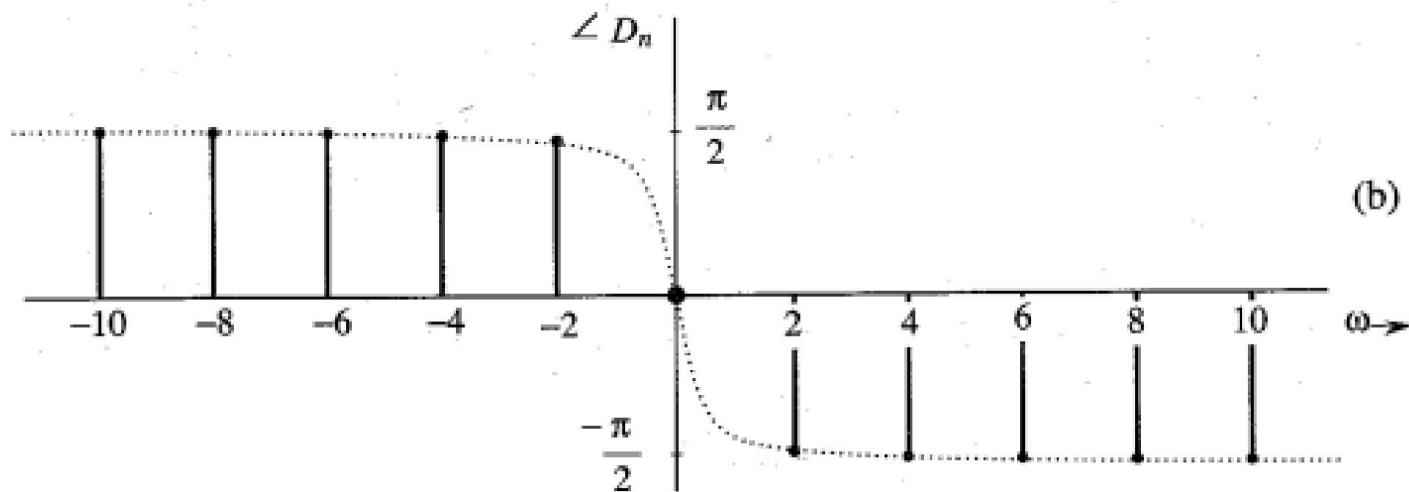
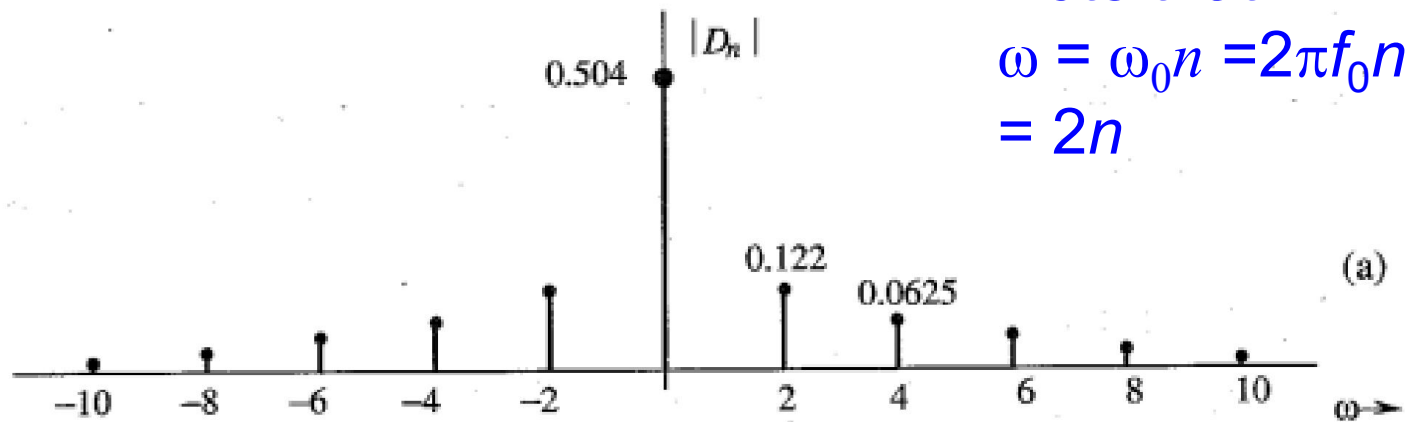
$$\angle D_n = \theta_n \quad \text{and} \quad \angle D_{-n} = -\theta_n$$

- $|D_n|$ are amplitudes and $\angle D_n$ are angles of exponential spectra
- $|D_n|$ vs. f is an even function
- $\angle D_n$ vs. f is an odd function

Conjugate Symmetry Property in Fourier Series of Periodic signals

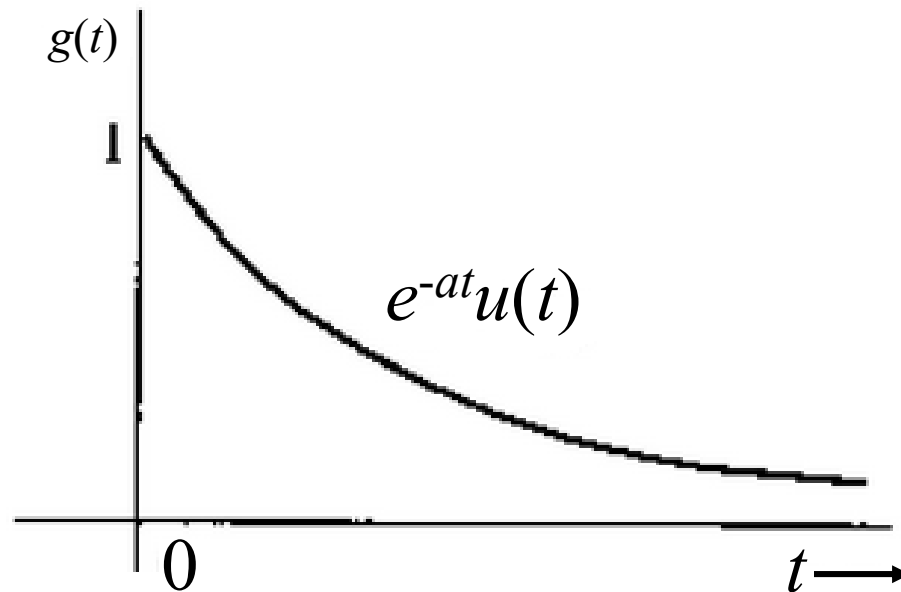
Revisited!

Note that
 $\omega = \omega_0 n = 2\pi f_0 n$
 $= 2n$



Fourier Transform Example

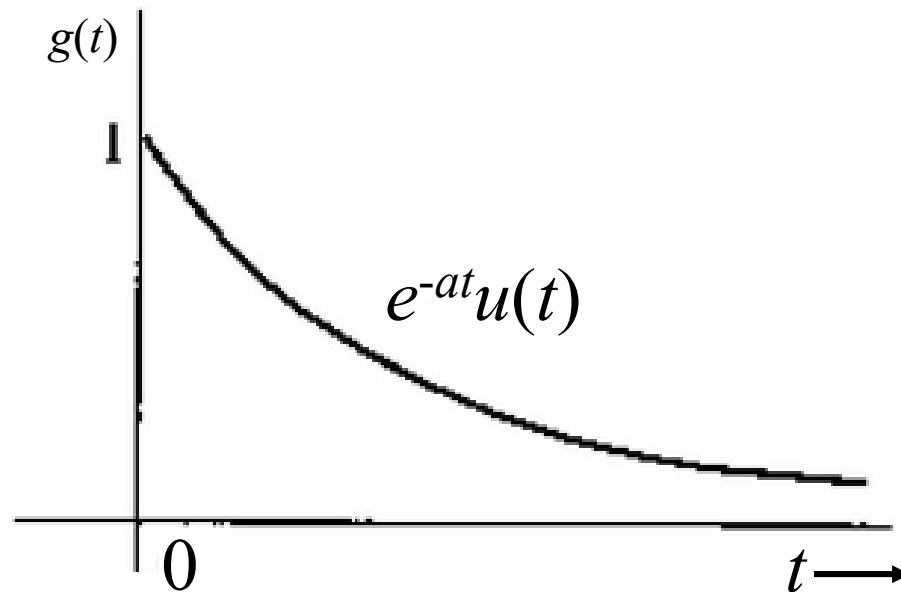
Find Fourier transform of $e^{-at}u(t)$



Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$G(f) = \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j2\pi ft} dt = \int_0^{\infty} e^{-at}e^{-j2\pi ft} dt$$

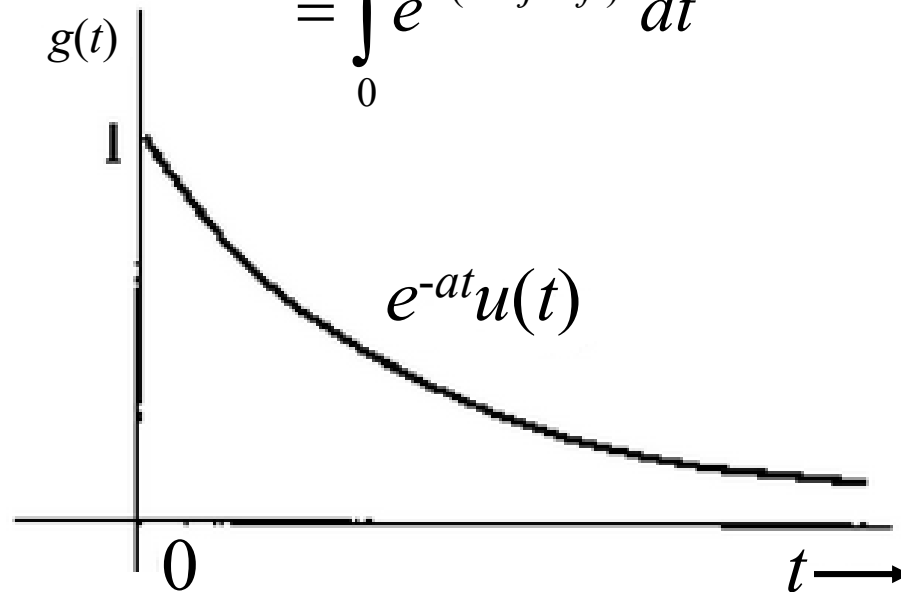


Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j2\pi ft} dt = \int_0^{\infty} e^{-at} e^{-j2\pi ft} dt$$

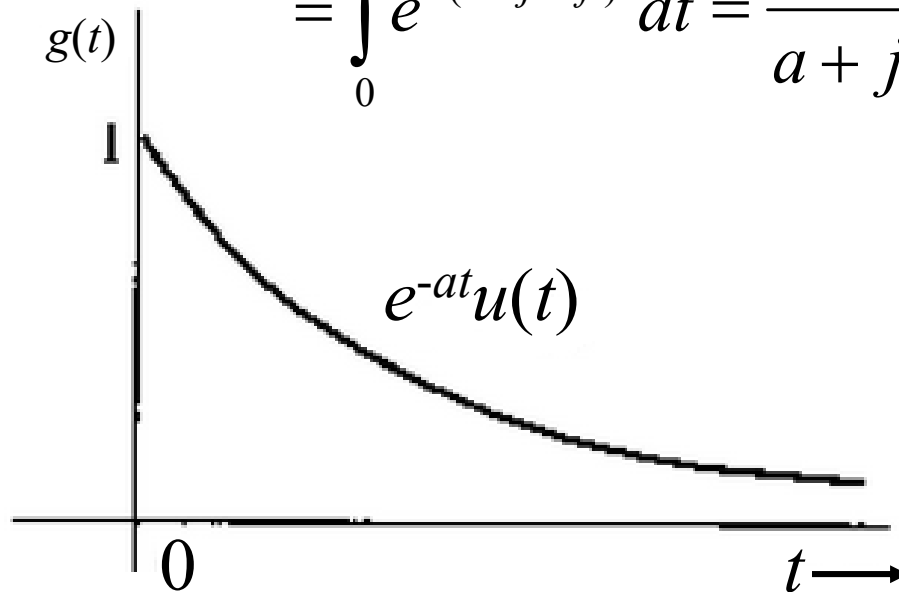
$$= \int_0^{\infty} e^{-(a+j2\pi f)t} dt$$



Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$\begin{aligned} G(f) &= \int_{-\infty}^{\infty} g(t) e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j2\pi ft} dt = \int_0^{\infty} e^{-at} e^{-j2\pi ft} dt \\ &= \int_0^{\infty} e^{-(a+j2\pi f)t} dt = \frac{-1}{a+j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^{\infty} \end{aligned}$$



Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$\begin{aligned} G(f) &= \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j2\pi f t} dt = \int_0^{\infty} e^{-at} e^{-j2\pi f t} dt \\ &= \int_0^{\infty} e^{-(a+j2\pi f)t} dt = \frac{-1}{a+j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^{\infty} \end{aligned}$$

Now, as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$\begin{aligned} G(f) &= \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t} dt = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-j2\pi f t} dt = \int_0^{\infty} e^{-at} e^{-j2\pi f t} dt \\ &= \int_0^{\infty} e^{-(a+j2\pi f)t} dt = \frac{-1}{a+j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^{\infty} \end{aligned}$$

Now, as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$\begin{aligned} G(f) &= \int_{-\infty}^{\infty} g(t)e^{-j2\pi ft} dt = \int_{-\infty}^{\infty} e^{-at}u(t)e^{-j2\pi ft} dt = \int_0^{\infty} e^{-at}u(t)e^{-j2\pi ft} dt \\ &= \int_0^{\infty} e^{-(a+j2\pi f)t} dt = \frac{-1}{a+j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^{\infty} \\ &= \frac{-1 \times -1}{a+j2\pi f} = \frac{1}{a+j\omega} \end{aligned}$$

Now, as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi ft} = 0$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$G(f) = \frac{1}{a + j\omega} = \frac{1}{a + j2\pi f}$$

$$G(f) = \frac{1}{\sqrt{a^2 + (2\pi f)^2}} e^{-j \tan^{-1} \left(\frac{2\pi f}{a} \right)}$$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

$$G(f) = \frac{1}{a + j\omega} = \frac{1}{a + j2\pi f}$$

$$G(f) = \frac{1}{\sqrt{a^2 + (2\pi f)^2}} e^{-j \tan^{-1} \left(\frac{2\pi f}{a} \right)}$$

Therefore,

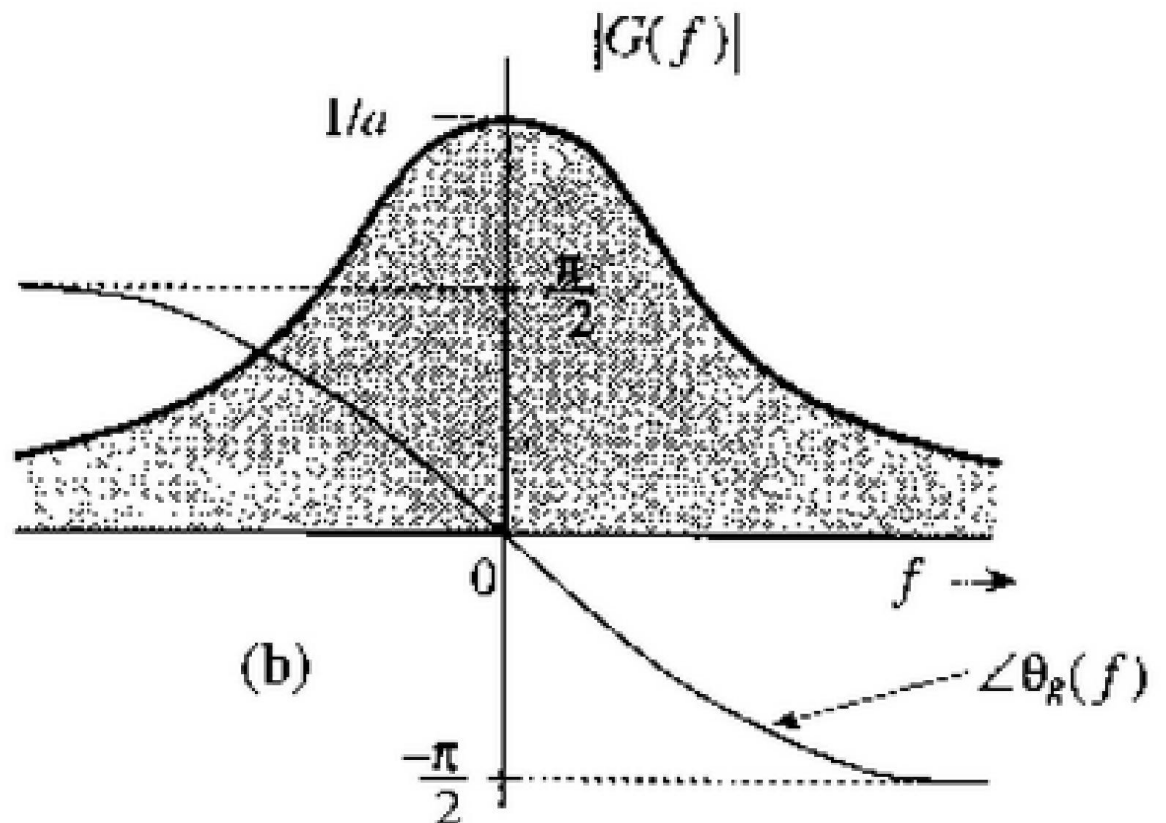
$$|G(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}} \quad \text{and} \quad \theta_g(f) = -\tan^{-1} \left(\frac{2\pi f}{a} \right)$$

Fourier Transform Example

Find Fourier transform of $e^{-at}u(t)$

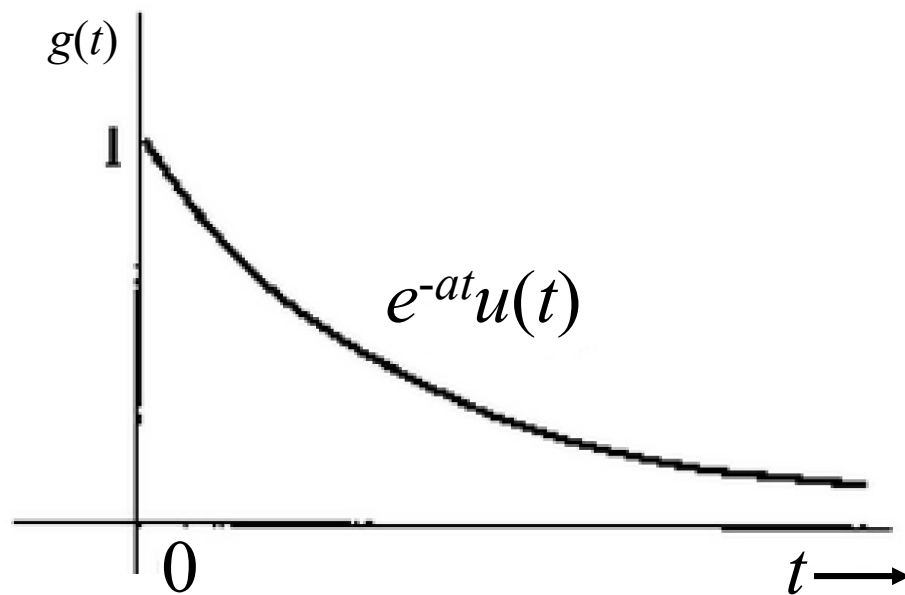
$$|G(f)| = \frac{1}{\sqrt{a^2 + (2\pi f)^2}}$$

$$\theta_g(f) = -\tan^{-1}\left(\frac{2\pi f}{a}\right)$$



Fourier Transform Example

Remember, we got this result for $a > 0$

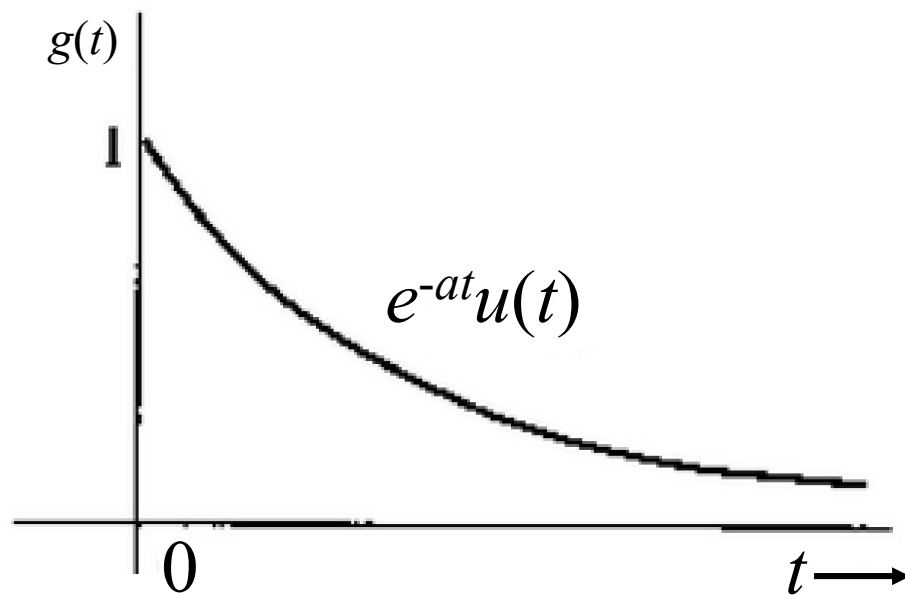


$$\begin{aligned} G(f) &= \frac{-1}{a + j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^\infty \\ &= \frac{1}{a + j2\pi f} \end{aligned}$$

Now as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Fourier Transform Example

What happens if result for $a < 0$?



$$\begin{aligned} G(f) &= \frac{-1}{a + j2\pi f} \left[e^{-(a+j2\pi f)t} \right]_0^\infty \\ &= \frac{1}{a + j2\pi f} \end{aligned}$$

Now as $t \rightarrow \infty$, $e^{-(a+j2\pi f)t} = e^{-at} e^{-j2\pi f t} = 0$ provided $a > 0$

Existence of Fourier Transform

Conditions

- Area under curve must be finite
- The function must be integrable

$$\int_{-\infty}^{\infty} |g(t)| dt < \infty$$

Dirichlet condition